

Autonomous Lecture $\rightarrow$ Controls part 4

State variable: $\bar{x}$

Observed Variable: $\bar{y}$

Let differential equation:

$$\dot{u} = au \rightarrow \text{solution: } u(t) = e^{at} u(0)$$

But if $\dot{\bar{x}} = A\bar{x} \Rightarrow \text{solution} \Rightarrow \bar{x}(t) = e^{At} \bar{x}(0)$

for $A \in \mathbb{R}^{n \times n}$
 $\bar{x}, \dot{\bar{x}} \in \mathbb{R}^n$

where $e^{At} = I + At + \frac{1}{2}A^2t^2 + \dots$

Remember Diagonalization? : let T and D such that

$$A = TDT^{-1}$$

$$At = T(Dt)T^{-1}$$

$$I + At \Rightarrow I + T(Dt)T^{-1}$$

$$I + At + \frac{A^2t^2}{2} = T T^{-1} + T(Dt)T^{-1} + \frac{1}{2} T D T^{-1} T D T^{-1} (t^2)$$

$$T D D T^{-1} \Rightarrow T D^2 T^{-1} t^2$$

$$A = TDT^{-1}$$

where $D = \begin{bmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ 0 & & \ddots \\ & & & \lambda_n \end{bmatrix}$

$T \rightarrow$ eigen-basis matrix.

$$e^{At} = T e^{Dt} T^{-1}$$

hence: $e^{At} = T e^{Dt} T^{-1}$

so: $\dot{\bar{x}} = A\bar{x} \xrightarrow{\text{has soln}} \bar{x} = e^{At} \bar{x}(0) = T e^{Dt} T^{-1} \bar{x}(0)$

where $e^{Dt} = I + Dt + \frac{1}{2}D^2t^2 + \dots$

But since $D \rightarrow$ diagonal matrix.

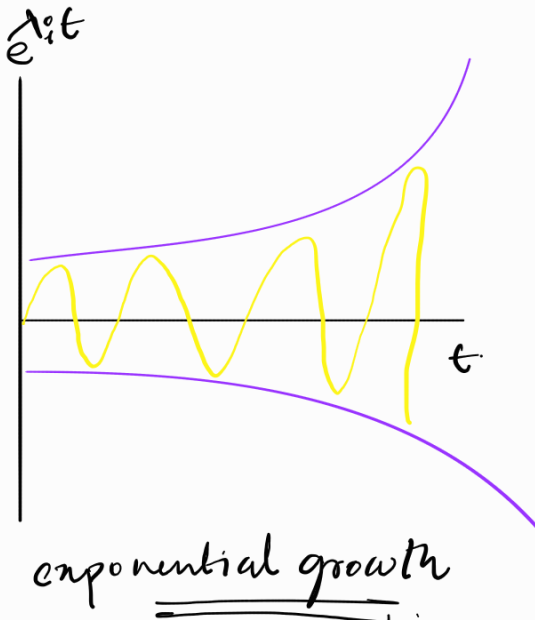
$$e^{Dt} = \begin{bmatrix} e^{\lambda_1 t} & & 0 \\ & e^{\lambda_2 t} & \\ 0 & & \ddots \\ & & & e^{\lambda_n t} \end{bmatrix}$$

why is this important?

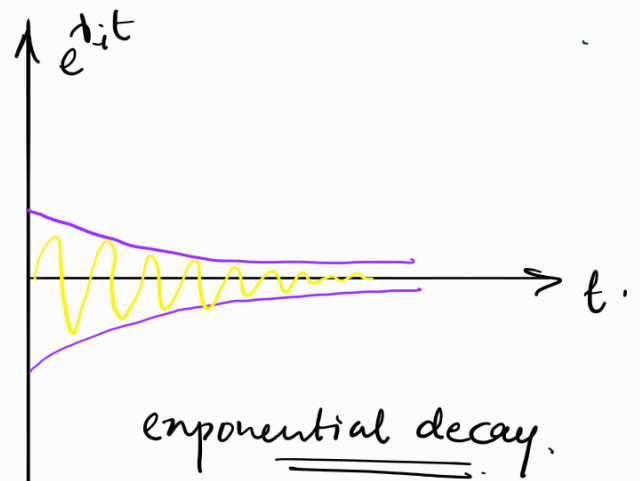
let $\lambda_i \in \mathbb{C}$ i.e. $\lambda_i = a_i + j b_i$

i.e. $\lambda_i = \underbrace{e^{a_i t}}_{\text{exponential component}} \underbrace{(\cos b_i t \pm j \sin b_i t)}_{\text{sinusoid component}}$

if $a_i > 0$



if $a_i < 0$



Which one is preferable?

our solution: $\bar{n}(t) = T e^{Dt} T^{-1} \bar{n}(0)$

$$= T \begin{bmatrix} e^{\lambda_1 t} & 0 \\ 0 & e^{\lambda_2 t} \\ 0 & \ddots \\ 0 & 0 & e^{\lambda_n t} \end{bmatrix} T^{-1} \bar{n}(0)$$

$\Rightarrow$ even a single $a_i > 0$

de-stabilizes our dynamics.

hence : for stable dynamics. $\text{Re}\{\lambda_i\} < 0$

$\forall i = 1, 2, \dots, n.$

⊛ Prateek has showed how to form state-space eqns. for dynamical systems.

Linearization : let $\dot{\bar{x}} = f(\bar{x}) \rightarrow$ non-linear function.

but we want $\dot{\bar{x}} = A\bar{x} \rightarrow$ how?

⊛ Step ① find $\bar{x} = \bar{x}_0$ such that

$$\begin{aligned} \dot{\bar{x}} &= \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \rightarrow \left. \begin{aligned} f(x_0) &= 0 \\ \text{i.e. } \dot{\bar{x}} \Big|_{\bar{x}=\bar{x}_0} &= 0 \end{aligned} \right\} \end{aligned}$$

⊛ Step ② linearize about x_0 $\Rightarrow \frac{Df}{Dx} \Big|_{\bar{x}=\bar{x}_0}$ (jacobian) $\epsilon > 0$

put $\dot{\bar{x}} = f(x)$

$\bar{x} \in V_\epsilon(x_0)$

$$= \left[\cancel{f(x_0)}_0 + \frac{Df}{Dx} \Big|_{\bar{x}=\bar{x}_0} (\bar{x} - x_0) + \frac{D^2 f}{Dx^2} \Big|_{\bar{x}=\bar{x}_0} (\bar{x} - x_0)^2 \right]$$

ignore higher order terms.

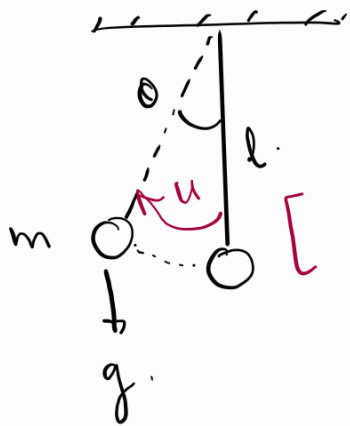
$$\dot{\bar{x}} = \frac{Df}{Dx} \Big|_{\bar{x}=\bar{x}_0} (\bar{x} - x_0) \Delta x.$$

$\rightarrow$ Then just shift origin to x_0 .

$$\text{then } \dot{\bar{x}} = \frac{Df}{Dx} \Big|_{\bar{x}=\bar{x}_0} \bar{x} ; \text{ let } \frac{Df}{Dx} \Big|_{\bar{x}=\bar{x}_0} = A = \text{jacobian.}$$

- Caveats:
- ① only valid in neighbourhood of x_0 .
 - ② only valid if eigenvalues of A are hyperbolic about x_0 .
- (Hartmann-Grobman)
- ie all eigenvalues of $\frac{DT}{Dx}|_{\bar{x}=x_0}$ have $\text{Re} < 0$.

Eg. Simple Pendulum:



$$\ddot{\theta} = -\frac{g}{L} \sin \theta - \underbrace{b \dot{\theta}}_{\text{friction or damping}}$$

let $\theta \rightarrow x_1$
 $\dot{\theta} \rightarrow x_2 \Rightarrow \bar{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix}$

$$\dot{\bar{x}} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} \dot{\theta} \\ \ddot{\theta} \end{bmatrix} = \begin{bmatrix} \dot{\theta} \\ -\frac{g}{L} \sin \theta - b \dot{\theta} \end{bmatrix}$$

let $\frac{g}{L} = 1$ (for convenience).

$\Rightarrow$ linearization: $\bar{\dot{x}} = 0$

$$\begin{bmatrix} \dot{\theta} \\ \ddot{\theta} \end{bmatrix} = 0 \Rightarrow \begin{cases} \dot{\theta} = 0 \\ \ddot{\theta} = 0 \end{cases} \Rightarrow \underbrace{-\sin \theta - b \dot{\theta}}_{=0} = 0 \Rightarrow \begin{cases} \dot{\theta} = 0 \\ \theta = 0 \text{ or } \pi \end{cases}$$

Compute jacobian:

$$\frac{DT}{Dx} = \begin{bmatrix} 0 & 1 \\ -\cos x_1 & -b \end{bmatrix} \Rightarrow \begin{bmatrix} 0 & 1 \\ -1 & -b \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & -b \end{bmatrix}$$

about $x = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$$\dot{\bar{x}} = \begin{bmatrix} 0 & 1 \\ -1 & -b \end{bmatrix} (\bar{x} - \begin{bmatrix} 0 \\ 0 \end{bmatrix})$$



$\rightarrow$ two points for linearizⁿ.

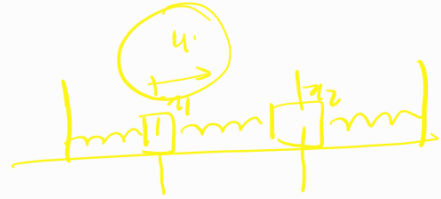
$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} \text{ or } \begin{bmatrix} 0 \\ \pi \end{bmatrix}$$



[Inverted pendulum]

$$\dot{\bar{x}} = \begin{bmatrix} 0 & 1 \\ 1 & -b \end{bmatrix} (\bar{x} - \begin{bmatrix} 0 \\ 1 \end{bmatrix}) \quad \left\{ \begin{array}{l} \text{only valid if } \underline{\underline{\operatorname{Re}(\operatorname{eig}(A)) > 0}} \\ \textcircled{9} \text{ Remember} \end{array} \right.$$

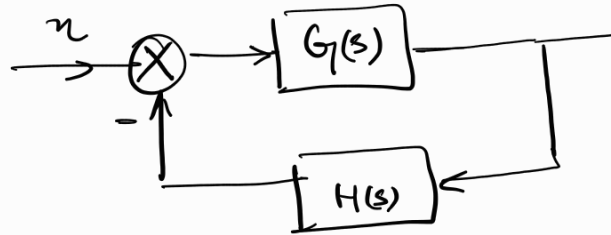
For some input u .



$\Rightarrow \dot{x} = Ax + Bu$ → Back to the state space equation.

Back to s-domain

Let control loop:



Open loop TF = $G(s)$

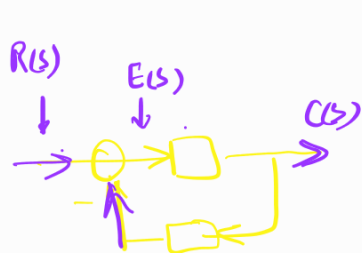
$$\text{C.L. TF} \Rightarrow \frac{G(s)}{1 + G(s)H(s)}$$

Let $H(s) = 1$ (for now) [unity feedback]

① First order systems: let $G(s) = \frac{1}{Ts}$

$$\text{C.L. TF} \Rightarrow \frac{\frac{1}{Ts}}{1 + \frac{1}{Ts}} \Rightarrow \frac{1}{Ts + 1} \quad (\text{for -ve unity feedback})$$

Step Response: give input = $\begin{cases} 1; t \geq 0 \\ 0; t < 0 \end{cases}$



$$\text{or } \begin{array}{c} \text{Step input} \\ t=0 \end{array} \Rightarrow \mathcal{L}\left\{ \frac{1}{s} \right\} \text{ for } t \geq 0$$

So: $\left(\frac{1}{s} \right) \left(\frac{1}{Ts+1} \right) \Rightarrow \frac{1}{Ts} \left(\frac{1}{s+\frac{1}{T}} \right)$ [Take inverse Laplace]

$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)}$

CL TF

partial fraction

$\frac{1}{T} \left(\frac{1}{s} - \frac{1}{s+\frac{1}{T}} \right) \xrightarrow{\mathcal{L}^{-1}} \left[1 - e^{-t/T} \right]$
 Step Response

Similarly put input = $\delta \rightarrow$ Dirac delta

$\rightarrow$ Impulse Response

$\frac{d}{dt} (\text{Step Response}) \Rightarrow \frac{1}{T} e^{-t/T}$



put input = $t \rightarrow$ Ramp input $\rightarrow$ gives ramp response.

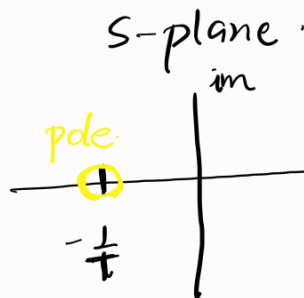
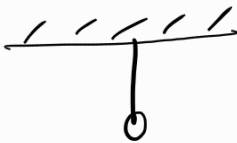


$\frac{1}{s^2}$

$\frac{d}{dt} (\text{Ramp Response}) = \text{Step Response}$

Further analysis of a first order system:

$\frac{C(s)}{R(s)} = \frac{1}{Ts+1} \Rightarrow$



for $T > 0$

[pole $\equiv$ eigenvalue]

(will not show why)

System is stable if: $\text{Re}(\text{pole}) < 0$

Any Generic 1st order system:

$\frac{K}{s+a} \Rightarrow$

$\frac{K}{a} \left(\frac{s}{\frac{s}{a} + 1} \right)$

$\frac{1}{a} \rightarrow T$

$\frac{K}{a} \rightarrow K' \rightarrow \text{const. coeff}^t$

$\rightarrow$ find Step Response

Second order system:

$$G(s) \xrightarrow{ut} \frac{\omega_n^2}{s(s+2\zeta\omega_n)} \xrightarrow{\text{poles}} \begin{array}{c} \text{Im} \\ -2\zeta\omega_n \quad | \quad 0 \\ \text{Re} \end{array}$$

$$\frac{G(s)}{1+G(s)} = \frac{C(s)}{R(s)} \Rightarrow \frac{\omega_n^2}{\omega_n^2 + 2\zeta\omega_n s + s^2} \rightarrow \text{poles} \quad \omega_n(-\zeta \pm \sqrt{\zeta^2 - 1})$$

Conditions

① if $\zeta = 1$: Critically damped; poles: $-\omega_n$

② if $\zeta < 1$: under damped - case: $\omega_n(-\zeta \pm \sqrt{\zeta^2 - 1})$

$$\Rightarrow -\sigma \pm j\omega_d \quad (\omega_d = \omega_n\sqrt{1-\zeta^2}) \\ (\sigma = \omega_n\zeta)$$

③ if $\zeta > 1$: Over damped case $\Rightarrow$

$$\omega_n(-\zeta \pm \sqrt{\zeta^2 - 1})$$

[Real Roots]

① Underdamped Step Response : $1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \left[\cos(\omega_d t) + \frac{\zeta}{\sqrt{1-\zeta^2}} \sin(\omega_d t) \right]$

$\swarrow$ How?

take. $\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)} \Rightarrow C(s) = \frac{R(s)}{1/s} \cdot \frac{G(s)}{1+G(s)} = \frac{\omega_n^2}{\omega_n^2 + 2\zeta\omega_n s + s^2}$

$\Rightarrow$ then $\bar{x}^{-1}$

$$1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \left[\cos(\omega_d t) + \frac{\zeta}{\sqrt{1-\zeta^2}} \sin(\omega_d t) \right]$$

$\underbrace{\hspace{10em}}_{\text{Exponential component}} \quad \underbrace{\hspace{10em}}_{\text{Sinusoidal Component}}$

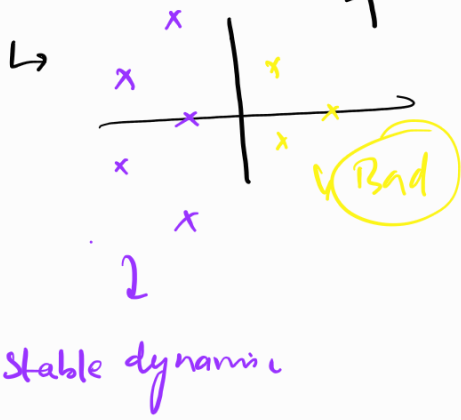
Response $\rightarrow$



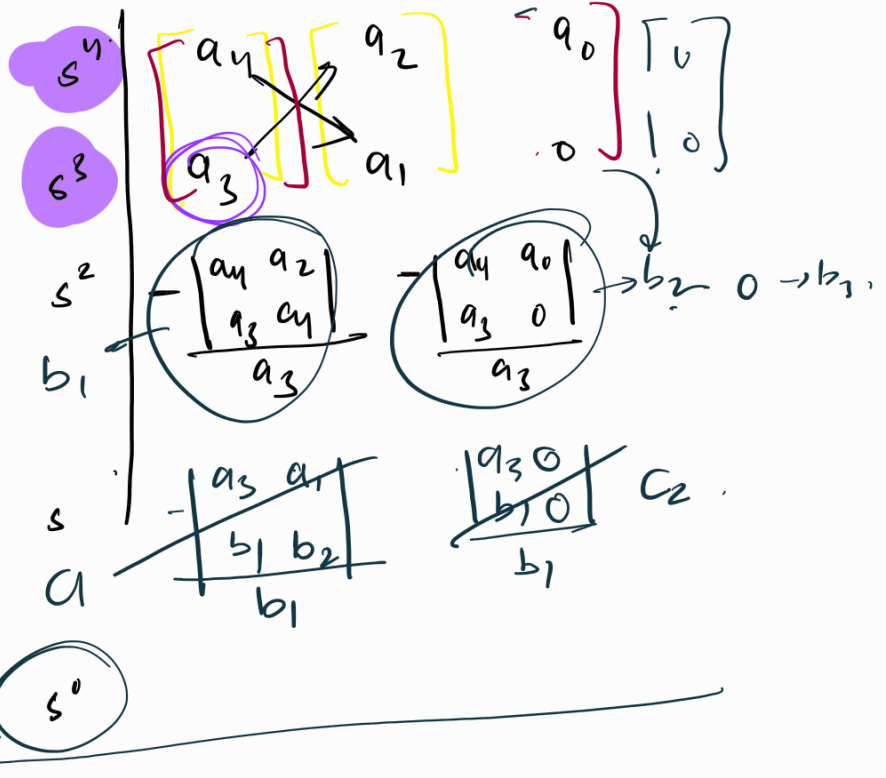
Stability

Routh-Hurwitz Criteria

s-plane

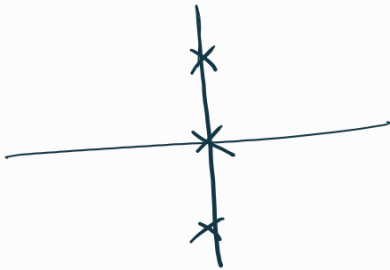


$$\frac{C(s)}{R(s)} = \frac{N(s)}{D(s)} = \frac{a_4 s^4 + a_3 s^3 + a_2 s^2 + a_1 s + a_0}{...}$$



s^4 $a_4 +$
 s^3 $a_3 +$
 s^2 $0 -$
 s^1 $c_1 +$
 s^0 $d_1 -$

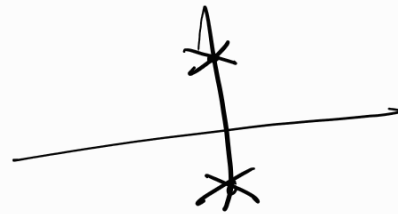
3 pole on the RHP



$$\begin{array}{c|c} s^5 & a_1 \\ s^4 & \varepsilon \\ s^3 & a_2 \end{array}$$

$$a_1 a_2 < 0$$

$$\begin{array}{c|c} s^4 & a_1 \quad a_2 \quad a_3 \\ s^3 & 3a_1 \quad a_2 \end{array} \rightarrow \frac{d}{ds}$$



Construct $\rightarrow$ auxiliary polynomial

$$P(s) = a_1 s^3 + a_2 s^1 \rightarrow \frac{dP}{ds} \Rightarrow 3a_1 s^2 + a_2$$

Routh Hurwitz criteria expl

Norman S. Nise → optimal control

↓ Steven. Brunton → Controls Bootcamp.